

CSS Gradients

What is a Gradient?

A gradient is a smooth transition from one color to another color.

Instead of using only one solid color, a gradient allows us to combine two or more colors and create a smooth visual effect.

For example:

Red → Orange → Yellow

The colors do not change suddenly. They gradually blend into each other.

Why Do We Use Gradients?

Gradients are commonly used to make web pages more attractive.

They can be used for:

- Backgrounds
- Buttons
- Cards
- Headers
- Banners
- Sections
- Navigation areas
- Text effects
- Decorative elements

Example:

A solid background:

BLUE

A gradient background:

BLUE → PURPLE → PINK

Gradients are created using CSS.

Types of CSS Gradients

CSS mainly provides two commonly used gradient types:

1. Linear Gradient
2. Radial Gradient

Linear Gradient

Definition

A linear gradient creates a smooth transition between colors along a straight line.

The colors move from one direction to another.

Simple idea:

Color 1
↓
↓
↓
Color 2

For example:

Blue
↓
↓
↓
Purple

The transition happens in a straight direction.

Syntax of Linear Gradient

The basic syntax is:

```
linear-gradient(direction, color1, color2, ...)
```

Example:

```
linear-gradient(to right, red, blue)
```

Here:

- `linear-gradient()` is the CSS gradient function.
- `to right` specifies the direction.
- `red` is the first color.
- `blue` is the second color.

Using Linear Gradient as a Background

A gradient is commonly applied using the `background` property.

Example:

```
.box {  
  background: linear-gradient(to right, red, blue);  
}
```

This creates a gradient that starts with red on the left and gradually changes to blue on the right.

Default Direction of Linear Gradient

If we do not specify a direction, the default direction is from top to bottom.

Example:

```
.box {  
  background: linear-gradient(red, blue);  
}
```

The result is approximately:

Red
↓
↓
↓
Blue

So:

top → red
bottom → blue

Linear Gradient from Left to Right

We can use:

to right

Example:

```
.box {  
  background: linear-gradient(to right, red, blue);  
}
```

Direction:

Red → → → → Blue

The gradient starts from the left and moves toward

Linear Gradient — Diagonal Directions

1. What is a Diagonal Linear Gradient?

A diagonal linear gradient is a linear gradient where the colors transition diagonally instead of moving only horizontally or vertically.

For example:

```
Color 1  
  ↘  
   ↘  
    ↘  
      Color 2
```

The gradient moves from one corner toward another corner.

2. Why Do We Use Diagonal Directions?

Diagonal gradients are useful for creating more dynamic backgrounds and visual effects.

For example:

Top Left
↘
↘
↘
Bottom Right

Instead of:

Left → → → → Right

or:

Top
↓
↓
Bottom

3. Diagonal Direction Using Keywords

CSS provides diagonal directions using two keywords.

The general syntax is:

```
linear-gradient(to corner, color1, color2)
```

The commonly used diagonal directions are:

```
to bottom right  
to bottom left  
to top right  
to top left
```

CSS Radial Gradient

What is a Radial Gradient?

A radial gradient is a CSS gradient in which colors spread outward from a central point.

Unlike a linear gradient, which moves in a straight line, a radial gradient spreads in a circular or elliptical pattern.

Simple idea:

Center
↓
Color 1
↓
Color 2
↓
Color 3

The colors gradually change as they move outward from the center.

Linear Gradient vs Radial Gradient

Linear Gradient

Colors change along a straight line.

Color 1 → → → → Color 2

Radial Gradient

Colors spread outward from a center point.

Color 2
↗ ↑ ↖
Color 2 ● Color 2
↘ ↓ ↙
Color 2

The ● represents the center of the gradient.

Syntax of Radial Gradient

The basic syntax is:

```
radial-gradient(shape, color1, color2, ...)
```

Example:

```
.box {  
  background: radial-gradient(red, blue);  
}
```

Here:

- `radial-gradient()` is the CSS gradient function.
- `red` is the first color.
- `blue` is the second color.

The gradient begins from the center and moves outward.

Radial Gradient with Three Colors

A radial gradient can contain more than two colors.

Example:

```
.box {  
  background: radial-gradient(red, yellow, blue);  
}
```

The colors transition like:

Red
↓
Yellow
↓
Blue

The first color appears near the center, and the other colors spread outward.

Radial Gradient Shape

Radial gradients can have different shapes.

The two important shapes are:

1. Circle
2. Ellipse

Circle

A circular radial gradient spreads equally in all directions.

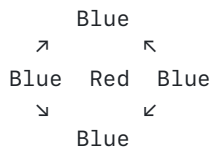
Syntax:

```
radial-gradient(circle, color1, color2)
```

Example:

```
.box {  
  background: radial-gradient(circle, red, blue);  
}
```

Conceptually:



Blue
↗ ↖
Blue Red Blue
↘ ↙
Blue

The gradient has a circular shape.

Ellipse

An ellipse is like a stretched circle.

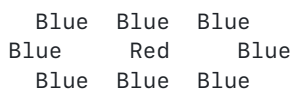
Syntax:

```
radial-gradient(ellipse, color1, color2)
```

Example:

```
.box {  
  background: radial-gradient(ellipse, red, blue);  
}
```

Conceptually:



Blue Blue Blue
Blue Red Blue
Blue Blue Blue

The gradient can stretch horizontally or vertically depending on the element.

Radial Gradient with Three Colors

Example:

```
.box {
  width: 300px;
  height: 200px;
  background: radial-gradient(
    red,
    yellow,
    blue
  );
}
```

The color transition is:

```
Center
↓
Red
↓
Yellow
↓
Blue
↓
Outside
```